Unit-3
Lecture-1

Born’s interpretation of the wave function,

Probability density and Normalization,

Born's interpretation of the wave function

- ψ is a wave function
- $|\psi|^2$ does not measure the particle density at any point and gives the probability of finding the particle at that point at any given moment.
- $|\psi|^2 = \psi \psi^*$
- More exactly, the probability of the particle being present in a volume dxdydz is $|\psi|^2$ dxdydz
- $|\psi|^2 dxdydz = 1$

  The wave function ψ satisfying this requirement is said to be normalized.
- $|\psi(x)|^2 dx = 1$ one dimensional Normalization wave equations
- $|\psi(x)|^2 dxdy = 1$ Two dimensional Normalization wave equations
- $|\psi(x)|^2 dxdydz = 1$ Three dimensional Normalization wave equations

Probability of finding of particle ⇒

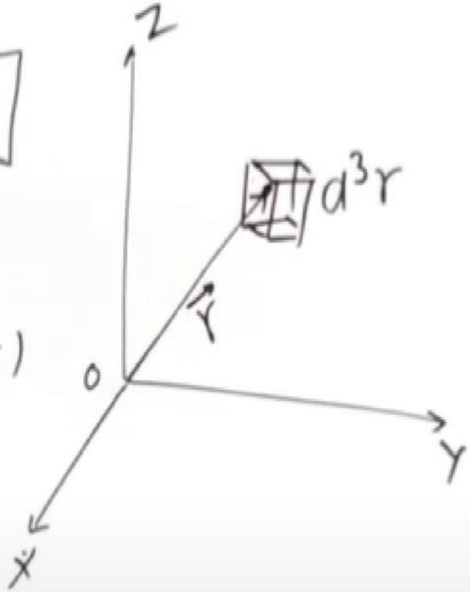
let wave funct^h of particle in $\psi(\vec{r},t)$, then probability of finding of particle in volume element $d^3r / d\tau / dV$ at time 't'

Max Born

- Probability of finding of particle
- Position Probability density
- Normalization cond^h
- Square integrable wave funct^h.

In 3-D

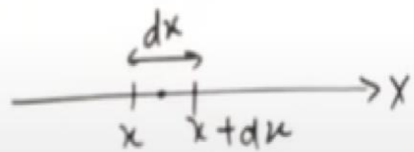
$$\left[P(\bar{r},t)\, d^3r = \psi^*(\bar{r},t)\,\psi(\bar{r},t)\, d^3r \right]$$



$\psi(\bar{r},t) \rightarrow$ wave funct^h

$\psi^*(\bar{r},t) \rightarrow$ Complex conjugat of $\psi(\bar{r},t)$

In-1D ⇒



Probability of finding of particle in range x to $x+dx$ at time 't'

$$\left[P(x,t)\, dx = \psi^*(x,t)\,\psi(x,t)\, dx \right]$$

Position probability density :-

Position probability density is define as

$$\boxed{P(\hat{r},t) = \psi^*(\bar{r},t)\psi(\bar{r},t)} \quad \text{— 3-D}$$

$$\left[P(\bar{r},t) = |\psi(\bar{r},t)|^2\right]$$

1-D $\left[P(x,t) = \psi^*(x,t)\psi(x,t)\right]$

or $\left[P(x,t) = |\psi(x,t)|^2\right]$

• Normalization Cond^n $\Rightarrow$

Since we know probability of finding of particle somewhere at any time 't'. unity (one)

Mathematical it is written as

$$\left[\int_{-\infty}^{\infty} \underline{\psi^*(x,t)\,\psi(x,t)}\,dx = 1\right]$$

it is Normalization Cond^n

$$\left[\int_{-\infty}^{\infty} |\psi(x,t)|^2\,dx = 1\right]$$

In 3.D $\Rightarrow$

$$\left[\int_{\text{all space}} \psi^*(\bar{r},t)\,\psi(\bar{r},t)\,d^3r = 1\right]$$

$$\left[\int_{\text{all space}} |\psi(\bar{r},t)|^2\,d^3r = 1\right]$$

Square integrible wave functn:-

Question: The wave function of a particle position in space is given by ………………

que $\Rightarrow$ $\psi(x) = A \cdot e^{\frac{iP_0}{\hbar}x}$ $|x| < a$

Find A, $A \rightarrow$ Normalization Condn

$|x| < a \Rightarrow \begin{matrix} x > -a \\ x < a \end{matrix} \Rightarrow -a < x < a$

$$\psi(x) = \begin{cases} Ae^{\frac{iP_0x}{\hbar}} & -a < x < a \\ 0 & \text{otherwise.} \end{cases}$$

Solution

$\psi^*(x) = Ae^{\frac{-iP_0x}{\hbar}}$

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx = 1$$

$$\int_{-a}^{a} Ae^{-iP_0x/\hbar} \cdot Ae^{iP_0x/\hbar}dx = 1$$

$$A^2\int_{-a}^{+a} 1dx = 1 \Rightarrow A^2[x]_{-a}^{a} = 1$$

$A^2[a+a] = 1$

$A^2 \times 2a = 1$

$A^2 = 1/2a$

$\left[A = \frac{1}{\sqrt{2a}}\right]$

Normalised wave functn

$$\left[\psi(x) = \frac{1}{\sqrt{2a}}e^{iP_0x/\hbar}\right]$$